MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED, LOW-POWER SCHOTTKY TTL, FLIP-FLOPS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/371A, dated 8 August 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 28 and 29

FIGURE 5, NOTE 2, t_p: Delete "15 ns" and substitute "17 ns".

PAGE 35

FIGURE 5, NOTE 2: Delete "PRR is 50% of the clock PRR".

PAGE 36

FIGURE 5, NOTE 3: Delete "5%" and substitute "1%."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
5	5
6	6
33	Reprinted without change
34	34

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PAGE 49

TABLE III, device type 01, test numbers 45 through 54, V_{CC} column: Delete "5.0 V" and substitute "5.5 V".

TABLE III, device type 01, I_O, test numbers 49 through 52, Min and Max limits column: Delete "-30" and "-112" and substitute "10/".

TABLE III, device type 01, I_{CC}, test numbers 53 and 54, maximum limits column: Delete "40" and substitute "4.0".

TABLE III, Subgroup 7, Symbol column: Add "9/".

* TABLE III, device type 01, truth table tests, test number 63, 1Q, $1\overline{Q}$, $2\overline{Q}$, and 2Q columns: Delete "H", "L", "L", and "H" and substitute "L", "H", "H", and "L", respectively.

* TABLE III, device type 01, truth table tests, test number 79, 1Q, $1\overline{Q}$, $2\overline{Q}$, and 2Q columns: Delete "H", "L", "L", and "H" and substitute "L", "H", "H", and "L", respectively.

* TABLE III, device type 01, truth table tests, test number 81, 1Q, $1\overline{Q}$, $2\overline{Q}$, and 2Q columns: Delete "L", "H", "H", and "L" and substitute "H", "L", "L", and "H", respectively.

PAGE 51

Add new footnote 9/ as follows:

"9/ A = 3.0 V minimum; B = 0.0 V or GND."

TABLE III, add footnote 10/ as follows:

"

Test	Min/Max limits in mA for circuit		
	A	B	C
I_O	-20/-112	-30/-112	-30/-112

"

PAGE 53

TABLE III, device type 02, I_O, test numbers 57 through 60, Min and Max limits column: Delete "-30" and "-112" and substitute "11/".

TABLE III, Subgroup 7, Symbol column: Add "10/".

PAGE 54

TABLE III, device type 02, t_{PLH1}, test number 107, measured terminal column: Delete "$1\overline{CLR}$ to 1Q" and substitute "1CLR to 1Q".

TABLE III, device type 02, t_{PLH1}, test number 108, $1\overline{PR}$ column: Delete "3.5 V" and substitute "IN".

TABLE III, device type 02, t_{PHL1}, test number 111, measured terminal column: Delete "$1\overline{CLR}$ to $1\overline{Q}$" and substitute "1CLR to 1Q".

TABLE III, device type 02, t_{PHL1}, test number 112, measured terminal column: Delete "$1\overline{PR}$ to 1Q" and substitute "1PR to 1Q".

TABLE III, device type 02, t_{PHL1}, test number 113, measured terminal column: Delete "$2\overline{PR}$ to 2Q" and substitute "2PR to 2Q".

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TABLE III, device type 02, t_{PHL1}, test number 114, measured terminal column: Delete "$2\overline{CLR}$ to $2\overline{Q}$" and substitute "2CLR to 2Q".

TABLE III, device type 02, t_{PLH2}, test number 115, measured terminal column: Delete "1CLR to 1Q" and substitute "1CLK to 1Q".

TABLE III, device type 02, t_{PLH2}, test number 117, measured terminal column: Delete "2CLR to 2Q" and substitute "2CLK to 2Q".

TABLE III, device type 02, t_{PLH2}, test number 118, measured terminal column: Delete "2CLK to $2\overline{Q}$" and substitute "2CLK to 2Q".

TABLE III, device type 02, t_{PHL2}, test number 121, measured terminal column: Delete "2CLK to 2Q" and substitute "2CLK to 2Q".

TABLE III, device type 02, t_{PHL2}, test number 122, measured terminal column: Delete "2CLK to $2\overline{Q}$" and substitute "2CLK to 2Q".

PAGE 55

Add new footnote 10/ as follows:

"10/ A = 3.0 V minimum; B = 0.0 V or GND."

TABLE III, add footnote 11/ as follows:

"

Test	Min/Max limits in mA for circuit		
	A	B	C
I_O	-20/-112	-30/-112	-30/-112

"

PAGE 57

TABLE III, I_{IH5}, Max limits column: Delete "10" and substitute "20".

TABLE III, Subgroup 7, Symbol column: Add "10/".

PAGE 58

TABLE III, device type 03, t_{PLH1}, test number 99, measured terminal column: Delete "1CLR to $1\overline{Q}$" and substitute "1CLR to 1Q".

TABLE III, device type 03, t_{PLH1}, test number 100, measured terminal column: Delete "$2\overline{CLR}$ to 2Q" and substitute "2CLR to 2Q".

TABLE III, device type 03, t_{PLH1}, test number 101, measured terminal column: Delete "$2\overline{PR}$ to 2Q" and substitute "2PR to 2Q".

TABLE III, device type 03, t_{PHL1}, test number 104, $2\overline{Q}$ column: Delete "OUT".

TABLE III, device type 03, t_{PHL1}, test number 104, 2Q column: Add "OUT".

TABLE III, device type 03, t_{PHL1}, test number 104, measured terminal column: Delete "$2\overline{CLR}$ to $2\overline{Q}$" and substitute "2CLR to 2Q".

TABLE III, device type 03, t_{PHL1}, test number 105, 2Q column: Delete "OUT".

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TABLE III, device type 03, t_{PHL1}, test number 105, $2\overline{Q}$ column: Add "OUT".

TABLE III, device type 03, t_{PHL1}, test number 105, measured terminal column: Delete "$2\overline{PR}$ to 2Q" and substitute "2PR to 2Q".

TABLE III, device type 03, t_{PLH2}, test number 108, measured terminal column: Delete "2CLK to 2Q" and substitute "2CLK to 2Q".

TABLE III, device type 03, t_{PLH2}, test number 109, measured terminal column: Delete "2CLK to $2\overline{Q}$" and substitute "2CLK to 2Q".

TABLE III, device type 03, t_{PHL2}, test number 112, measured terminal column: Delete "2CLK to 2Q" and substitute "2CLK to 2Q".

TABLE III, device type 03, t_{PHL2}, test number 113, measured terminal column: Delete "2CLK to $2\overline{Q}$" and substitute "2CLK to 2Q".

PAGE 59

Add new footnote 10/ as follows:

"10/ A = 3.0 V minimum; B = 0.0 V or GND."

TABLE III, footnote 9/, I_O, circuit A: Delete "-30/-112" and substitute "-20/-112".

PAGE 60

TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, device type 04, V_{IC}, test number 25, 8D column: Add "-18 mA".

PAGE 61

TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, device type 04, I_O, test numbers 57 through 64, Min and Max limits column: Delete "-15" and "-110" and substitute "8/" (alters changes made by amendment 1).

TABLE III, I_{CCH}, Max Limits column: Delete "17" and substitute "18".

TABLE III, I_{CCL}, Max Limits column: Delete "23" and substitute "27".

TABLE III, I_{CCZ}, Max Limits column: Delete "27" and substitute "28".

TABLE III, Subgroup 7, Symbol column: Add "7/".

TABLE III, tests 84-90, column 11: Delete "L" and "H" and substitute "B" and "A" respectively.

TABLE III, f_{MAX}, Min Limits column: Delete "30" and substitute "35".

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PAGE 62

TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, t_{PLH2}, Max Limits column: Delete "12" and substitute "14".

TABLE III, t_{PLZ}, Min and Max Limits column: Delete "3" and "13" and substitute "2" and "12" respectively.

TABLE III, t_{PHZ}, Max Limits column: Delete "8" and substitute "10".

PAGE 63

TABLE III, device type 04, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, t_{PLH2}, Max Limits column: Delete "15" and substitute "18".

TABLE III, t_{PHL2}, Max Limits column: Delete "„" and substitute "17".

TABLE III, t_{PLZ}, Min and Max Limits column: Delete "3" and "15" and substitute "2" and "18" respectively.

TABLE III, t_{PHZ}, Max Limits column: Delete "10" and substitute "12".

Add new footnote 7/ as follows:

"7/ A = 3.0 V minimum; B = 0.0 V or GND."

TABLE III, add footnote 8/ as follows:

"

Test	Min/Max limits in mA for circuit		
	A	B	C
I_O	-20/-112	-30/-112	-30/-112

"

PAGE 64

TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

PAGE 65

TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

TABLE III, Subgroup 7, Symbol column: Add "7/".

PAGE 66

TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

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PAGE 67

TABLE III, device type 05, top of the page, delete "Terminal conditions (pins not desinated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)".

Add new footnote 7/ as follows:

"7/ A = 3.0 V minimum; B = 0.0 V or GND."

PAGE 70

TABLE III, Subgroup 7, Symbol column: Add "8/".

PAGE 72

Add new footnote 8/ as follows:

"8/ A = 3.0 V minimum; B = 0.0 V or GND."

PAGE 75

TABLE III, Subgroup 7, Symbol column: Add "8/".

PAGE 77

Add new footnote 8/ as follows:

"8/ A = 3.0 V minimum; B = 0.0 V or GND."

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The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1365)

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits		Unit
					Min	Max	
High-level input current at CLK or D	I_{IH6}	V_{CC} = 5.5 V V_{IN} = 7.0 V		01		100	µA
High-level input current at J, K, or $\overline{K}$	I_{IH7}			02, 03		100	µA
High-level input current at any input	I_{IH8}			04, 05, 06, 07			
High-level input current at $\overline{PR}$ or $\overline{CLR}$	I_{IH9}			01,02, 03		200	µA
High-level input current at CLK	I_{IH10}			02		400	µA
				03		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V V_O = 2.25 V		01, 02, 03, 04	-20	-112	mA
				05, 06, 07	-15	-110	mA
Output current, high level, outputs OFF	I_{OZH}	V_{CC} = 5.5 V V_O = 2.7 V		04, 05, 06, 07		20	µA
Output current, low level, outputs OFF	I_{OZL}	V_{CC} = 5.5 V V_O = 0.4 V		04, 05, 06, 07		-20	µA
Supply current 2/	I_{CC}	V_{CC} = 5.5 V V_{IN} = 0 V		01, 02		4.0	mA
				03		4.5	mA
Supply current, outputs high	I_{CCH}	V_{CC} = 5.5 V	V_{IN} = 5.0 V	05		17	mA
				04		18	
			V_{IN} = 0 V	06		15	
				07		21	
Supply current, outputs low	I_{CCL}	V_{CC} = 5.5 V	V_{IN} = 0 V	04		27	mA
				06		29	
			V_{IN} = 5.0 V	05		23	
				07		29	
Supply current, outputs disabled	I_{CCZ}	V_{CC} = 5.5 V V_{OC} = 5.0 V		05		27	mA
				04		28	
				06, 07		31	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

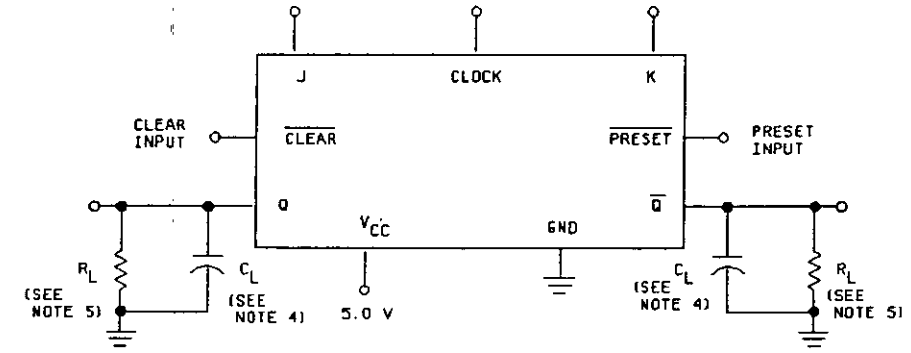
Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Maximum clock frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 500Ω	01,02,04, 05, 06, 07	30		MHz
			03	25		
Propagation delay time low to high level, CLR or PR to output	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 500Ω	01, 02	3	15	ns
			03	3	20	
Propagation delay time high to low, CLR or PR to output	t_{PHL1}		01, 02	5	17	
			03	4	22	
			05, 07	6	22	
Propagation delay time low to high level, CLK to output	t_{PLH2}		01, 02	5	18	
			03	3	18	
			05,06,07	4	15	
			04	4	18	
Propagation delay time high to low level, CLK to output	t_{PHL2}		01, 02	5	20	
			03	5	23	
			04, 05 06, 07	4	15	
Propagation delay time output control to low level output	t_{PZL}		04, 05, 06, 07	4	21	
Propagation delay time output control to high level output	t_{PZH}		04, 05, 06, 07	4	21	
Propagation delay time low level to output control	t_{PLZ}		05, 06, 07	3	15	
			04	2	15	
Propagation delay time high level to output control	t_{PHZ}		05, 06, 07	2	10	
			04	2	12	

1/ The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current, I_{OS}.

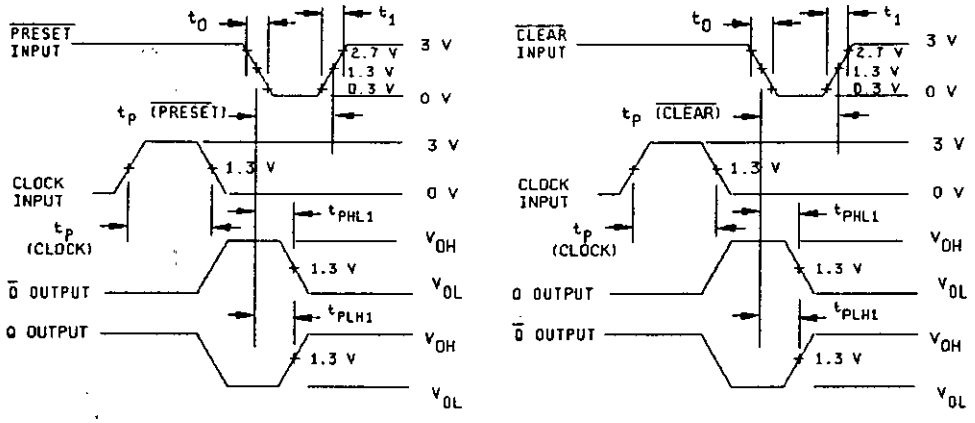
2/ I_{CC} is measured with outputs open with J, K, $\overline{K}$, or D, CLK and PR grounded; then with J, K, $\overline{K}$, or D, CLK and CLR grounded.

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of 8 August 1986

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TEST CIRCUIT



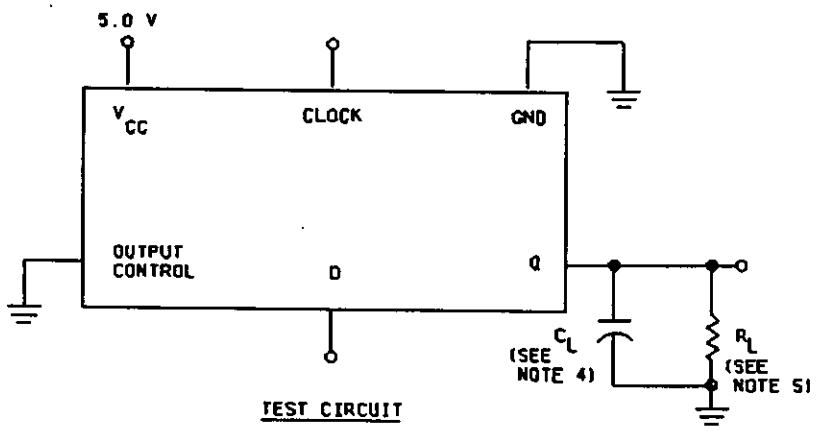
VOLTAGE WAVEFORMS

NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear and preset input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clear) = 15 ns; t_p (preset) = 15 ns; and PRR ≤ 1 MHz.
3. t_p (clock) = 20 ns minimum; PRR = clear or preset PRR.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. When testing clear to output switching, present inputs shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied (see table III).

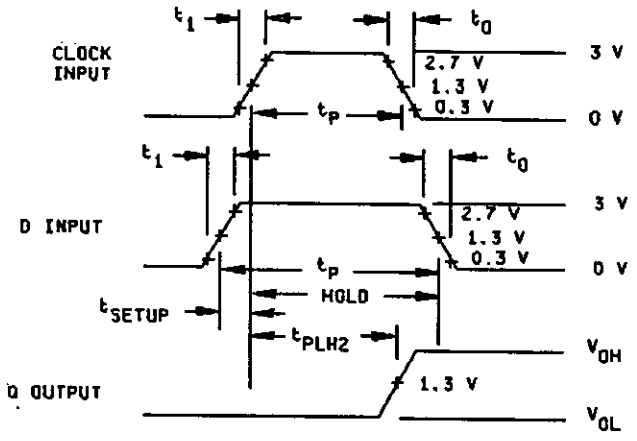
FIGURE 5. Clear and preset switching test circuit and waveforms (device type 03).

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR ≤ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{setup} = 15$ ns; $t_{hold} = 4$ ns; $t_p = 19$ ns; PRR is 50% of the clock PRR.
3. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz. The D input pulse shall be one half of the frequency of the clock and the D ↑ and ↓ shall be coincident with the clock ↓. $t_1 = t_0 = 6 \pm 1.5$ ns.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high level data) (device type 04).

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